

MSDS 7333 - Case Study 2: Analysis and Modeling of Diabetic Data

Tue Vu

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Contents

1	Introduction	2
1.1	Background	2
1.2	Dataset Description	2
1.3	Research Objectives	3
2	Exploratory Data Analysis	4
2.1	Data Overview and Preprocessing	4
2.2	Missing Value Analysis and Imputation	5
2.3	Categorical Variable Analysis	6
2.3.1	Key Categorical Variables	6
2.3.2	Excluded Categorical Variables	10
2.4	Numerical Variable Analysis	11
2.4.1	Correlation Analysis	12
2.4.2	Distribution Analysis	13
2.4.3	Relationship with Target Variable	14
2.5	Feature Engineering and Preparation for Modeling	15
3	Modeling with Logistic Regression	16
3.1	Model Selection	16
3.2	Data Partitioning	16
3.3	K-Fold Cross-Validation	16
3.4	Model Training	17
3.5	Model Evaluation	17
3.5.1	Performance Metrics	18
3.5.2	Test Set Results	18
3.6	Feature Importance Analysis	19
3.6.1	Coefficient Interpretation from Logistic Regression	19
3.6.2	Feature Importance from Random Forest	20
4	Conclusions	22
4.1	Summary of Findings	22
4.2	Methodological Limitations	23
4.3	Future Research Directions	23
4.4	Conclusion	23

Chapter 1

Introduction

1.1 Background

Diabetes represents a significant global health challenge, affecting millions of individuals worldwide and contributing substantially to healthcare expenditure. The condition is characterized by elevated blood glucose levels resulting from defects in insulin production, insulin action, or both. The complications associated with diabetes include cardiovascular disease, nephropathy, retinopathy, and neuropathy, significantly impacting patient quality of life and mortality rates.

Hospital readmissions among diabetic patients constitute a significant burden on healthcare systems, with implications for both quality of care and resource allocation. Readmissions within 30 days of discharge are particularly concerning as they may indicate inadequate initial treatment, premature discharge, or insufficient follow-up care. Understanding the factors associated with readmission patterns is essential for developing effective intervention strategies to improve patient outcomes and reduce healthcare costs.

1.2 Dataset Description

This analysis utilizes a comprehensive dataset containing records of diabetic patient encounters from 1999-2008 at 130 U.S. hospitals. The dataset comprises 101,766 observations across 50 variables, encompassing patient demographics, admission details, diagnoses, laboratory tests, and medication regimens. The target variable, "readmitted," categorizes patient outcomes into three classes: readmitted within 30 days (" <30 "), readmitted after 30 days (" >30 "), or not readmitted ("NO").

Key features in the dataset include:

- Demographic information ('age', 'race', 'gender', 'weight')
- Admission type, source, and discharge disposition ('admission_type_id', 'discharge_disposition_id', 'admission_source_id')
- Primary, secondary, and tertiary diagnoses ('diag_1', 'diag_2', 'diag_3')
- Laboratory tests ('A1Cresults', 'max_glu_serum')
- Medication changes during hospitalization ('num_medications')
- Diabetes medications (e.g., insulin, metformin, glyburide)

- Length of hospital stay ('time_in_hospital')
- Number of previous outpatient, inpatient, and emergency visits

1.3 Research Objectives

The primary objectives of this analysis are to:

1. Identify patterns and relationships within the diabetic patient data through comprehensive exploratory data analysis
2. There exists several missing values in the dataset, we need to handle and impute them appropriately
3. Develop a predictive model using logistic regression with k-fold cross-validation to estimate readmission probabilities
4. Evaluate the performance of the model and identify the most influential features affecting readmission outcomes
5. Determine the key factors associated with hospital readmission among diabetic patients

Through this investigation, we aim to contribute insights that may inform clinical decision-making and intervention strategies to reduce readmission rates among diabetic patients.

Chapter 2

Exploratory Data Analysis

2.1 Data Overview and Preprocessing

The exploratory data analysis began with an examination of the dataset structure and composition. The diabetic dataset comprises 101,766 patient encounters with 50 variables.

The distribution of the target variable "readmitted" reveals that 54,864 (53.9%) patients were not readmitted, 35,545 (34.9%) were readmitted after 30 days, and 11,357 (11.2%) were readmitted within 30 days (Figure 2.1). The target variable is imbalanced, which is a common issue in healthcare datasets.

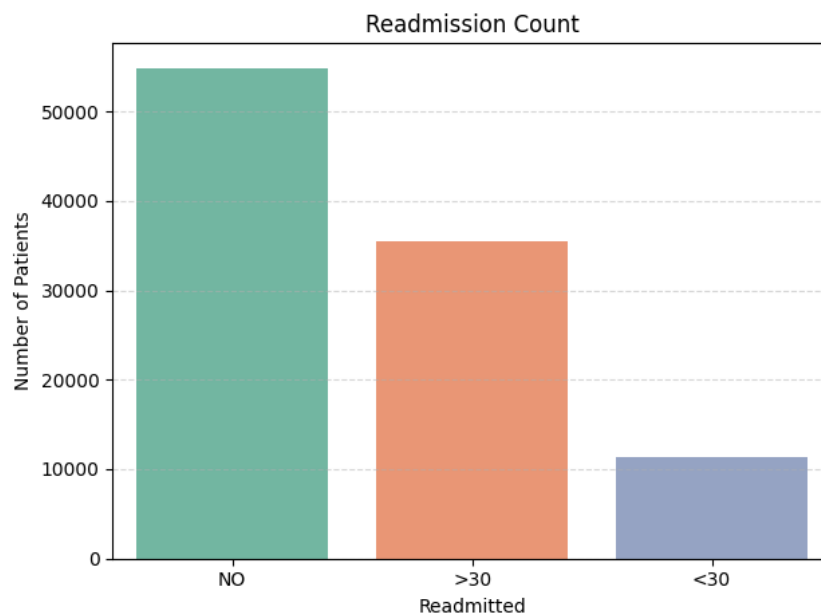


Figure 2.1: Distribution of patient readmission categories

Initial preprocessing steps included:

- Removal of identifier variables (encounter_id and patient_nbr)
- Replacement of "?" placeholders with NaN values for proper missing value handling

2.2 Missing Value Analysis and Imputation

A critical aspect of the preprocessing phase involved identifying and addressing missing values within the dataset. Nine variables exhibited missing values, with "weight" showing the highest proportion at 96.9%, followed by "max_glu_serum" and "A1Cresult" at 94.75% and 83.28% respectively.

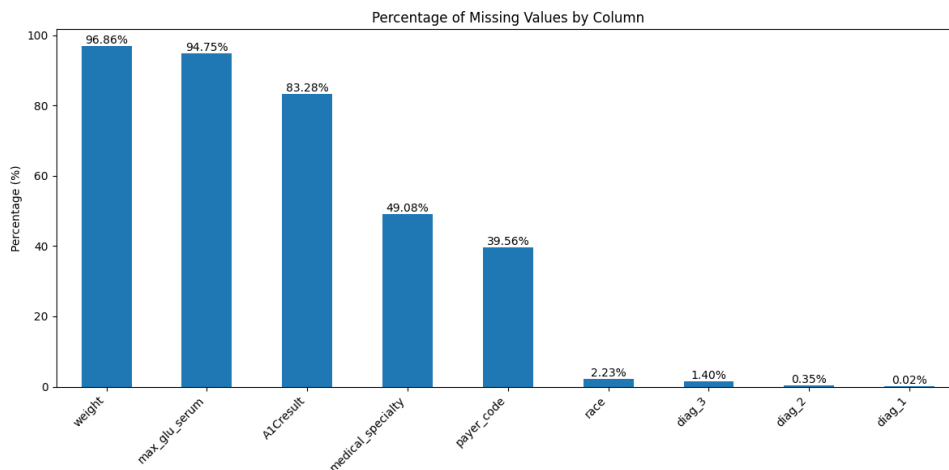


Figure 2.2: Percentage of missing values by variable

The following imputation strategies were implemented:

- Although "weight" is very important factor to identify the diabete condition, however, due to its extremely high percentage of missing values (96.9%) we dropped this variable.
- Despite its high percentage of missing values, due to small amount of categorical objects, the "max_glu_serum", "A1Cresult", and "medical_specialty", were imputed with the value "None"
- Similar to the above categorical variables, the "payer_code" and "race" variables were imputed with the value "None"
- For diagnosis variables: "diag_1", "diag_2", "diag_3", missing values were imputed with the value "None". Categories with less than 0.5% occurrence were merged into a new "Rest" category to reduce dimensionality and prevent overfitting. Due to its importance, the "diag_1" variable was considered more important than "diag_2", and "diag_2" was considered more important than "diag_3". If a patient has multiple diagnoses, the most important one was selected and put into the "diag" variable. The "Rest" category was created to handle the rare cases and thus has less importance ranking (after diag_3). The "None" category was created to handle the missing values and has the least importance (after "Rest"). This approach was used to handle the overfitting issue and reduce the dimensionality of the dataset.
- For "race", the missing values were imputed with "None", although it could be aligned with "Caucasian" or "AfricanAmerican" categories, due to high median values.
- For "age", the missing values were imputed with the median of the variable. ("age_mid")

2.3 Categorical Variable Analysis

The dataset contains numerous categorical variables representing patient demographics, diagnoses, medications, and hospital admission details. Categorical variables were categorized into two groups based on their distribution patterns and relationship with the target variable.

2.3.1 Key Categorical Variables

Based on distribution analysis and correlation with the target variable, the following categorical variables were identified as significant:

- Demographics: race, gender, payer_code (Figure 2.3)
- Medical details: max_glu_serum, A1Cresult, medical_specialty (Figure 2.4)
- Medications: metformin, glimepiride, glipizide, glyburide, pioglitazone, rosiglitazone (Figure 2.5)
- Treatment process: insulin,change, diabetesMed (Figure 2.6)
- Diagnosis: diag (Figure 2.7)
- Hospital factors: admission_type_id, discharge_disposition_id, admission_source_id (Figure 2.8)

Notable findings include:

- Race: Over 70% of patients are Caucasian, with distinct readmission patterns across racial groups (Figure 2.3)
- Admission type: Emergency intake (type 1) accounts for more than 50% of admissions (Figure 2.8)
- Discharge disposition: Approximately 60% of patients are discharged to home (Figure 2.8)
- Admission source: Nearly 60% of admissions originate from emergency rooms, followed by 30% from physician referrals (Figure 2.8)

Detailed information about the codes of Admission type, Discharge disposition, and Admission source are shown in Table 2.1, Table 2.2, and Table 2.3.

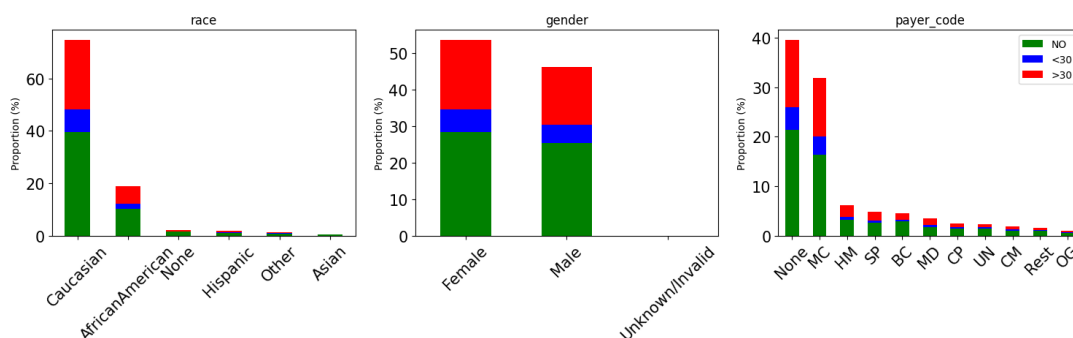


Figure 2.3: Distribution of readmission status across Demographics

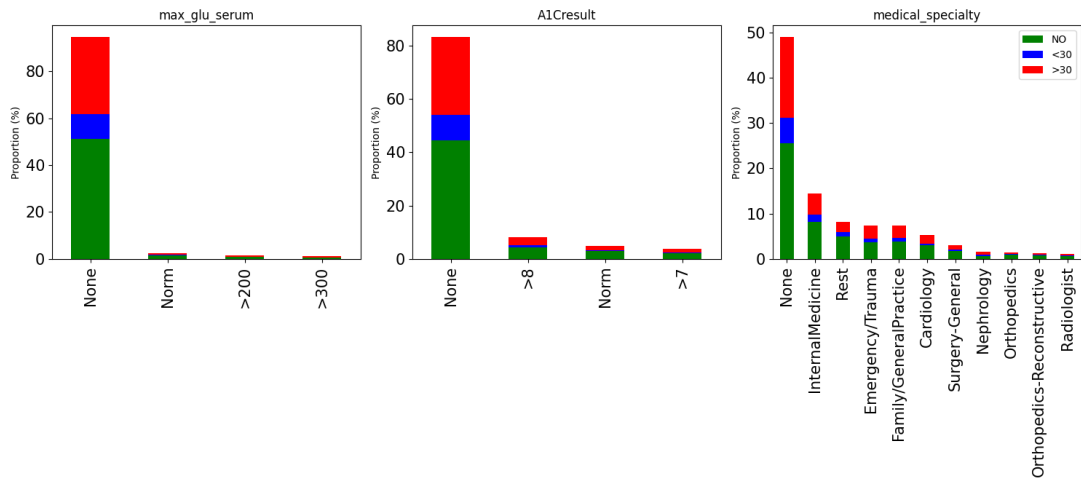


Figure 2.4: Distribution of readmission status across Medical details

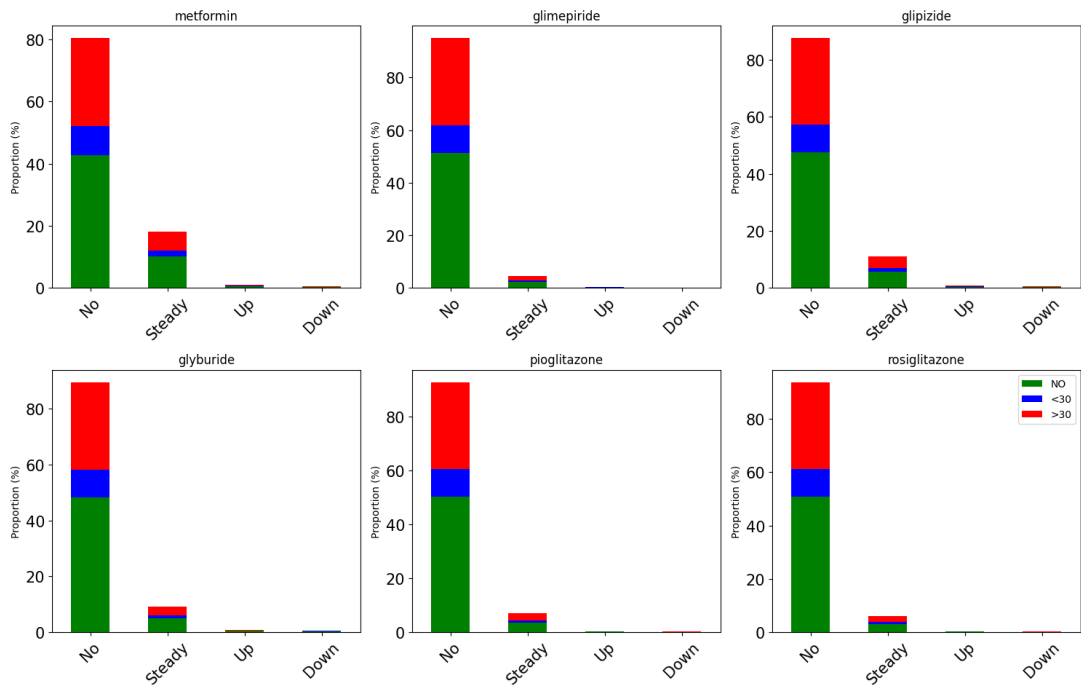


Figure 2.5: Distribution of readmission status across Medications

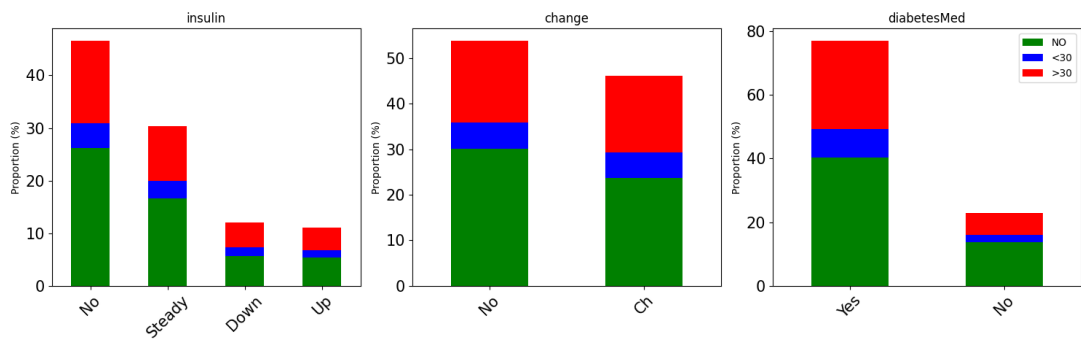


Figure 2.6: Distribution of readmission status across Treatment process

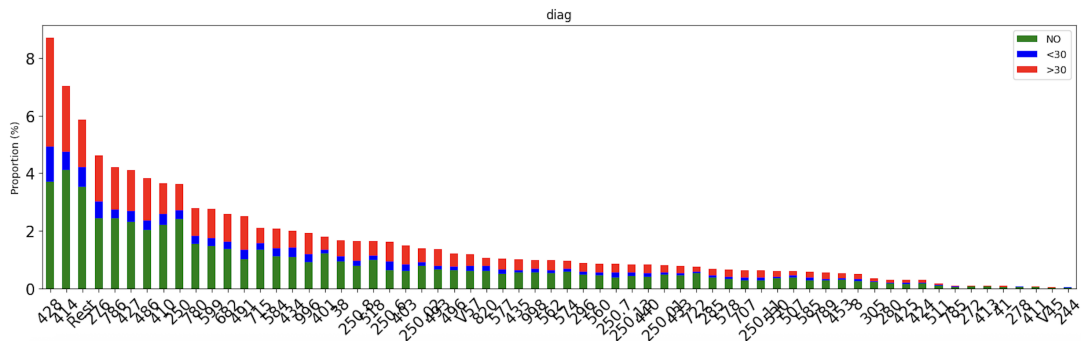


Figure 2.7: Distribution of readmission status across Diagnosis code

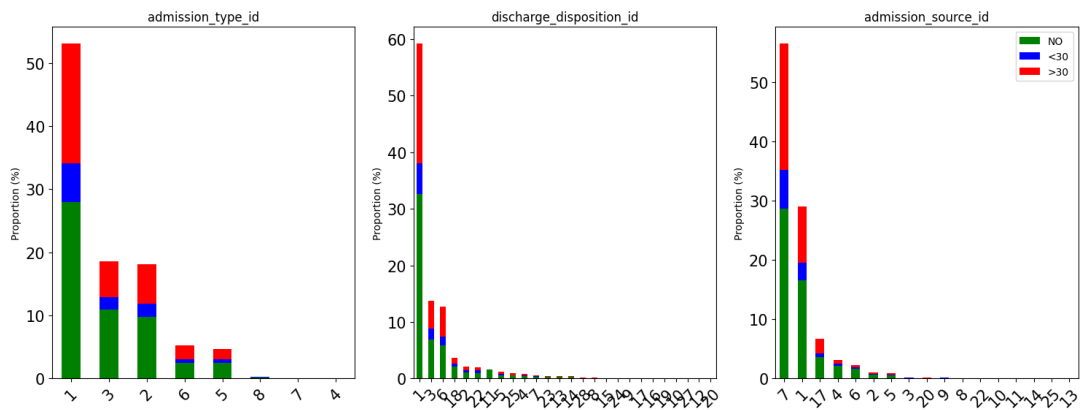


Figure 2.8: Distribution of readmission status across Hospital factors

Table 2.1: Admission types and their descriptions

admission_type_id	description
1	Emergency
2	Urgent
3	Elective
4	Newborn
5	Not Available
6	NULL
7	Trauma Center
8	Not Mapped

Table 2.2: Discharge Disposition Descriptions

discharge_disposition_id	description
1	Discharged to home
2	Discharged/transferred to another short term hospital
3	Discharged/transferred to SNF
4	Discharged/transferred to ICF
5	Discharged/transferred to another type of inpatient care institution
6	Discharged/transferred to home with home health service
7	Left AMA
8	Discharged/transferred to home under care of Home IV provider
9	Admitted as an inpatient to this hospital
10	Neonate discharged to another hospital for neonatal aftercare
11	Expired
12	Still patient or expected to return for outpatient services
13	Hospice / home
14	Hospice / medical facility
15	Discharged/transferred within this institution to Medicare approved swing bed
16	Discharged/transferred/referred another institution for outpatient services
17	Discharged/transferred/referred to this institution for outpatient services
18	NULL
19	Expired at home. Medicaid only, hospice.
20	Expired in a medical facility. Medicaid only, hospice.
21	Expired, place unknown. Medicaid only, hospice.
22	Discharged/transferred to another rehab fac including rehab units of a hospital.
23	Discharged/transferred to a long term care hospital.
24	Discharged/transferred to a nursing facility certified under Medicaid but not certified under Medicare.
25	Not Mapped
26	Unknown/Invalid
27	Discharged/transferred to a federal health care facility.
28	Discharged/transferred/referred to a psychiatric hospital of psychiatric distinct part unit of a hospital
29	Discharged/transferred to a Critical Access Hospital (CAH).
30	Discharged/transferred to another Type of Health Care Institution not Defined Elsewhere

Table 2.3: Admission Source Descriptions

admission_source_id	description
1	Physician Referral
2	Clinic Referral
3	HMO Referral
4	Transfer from a hospital
5	Transfer from a Skilled Nursing Facility (SNF)
6	Transfer from another health care facility
7	Emergency Room
8	Court/Law Enforcement
9	Not Available
10	Transfer from critical access hospital
11	Normal Delivery
12	Premature Delivery
13	Sick Baby
14	Extramural Birth
15	Not Available
17	NULL
18	Transfer From Another Home Health Agency
19	Readmission to Same Home Health Agency
20	Not Mapped
21	Unknown/Invalid
22	Transfer from hospital inpt/same fac reslt in a sep claim
23	Born inside this hospital
24	Born outside this hospital
25	Transfer from Ambulatory Surgery Center
26	Transfer from Hospice

2.3.2 Excluded Categorical Variables

Several medication variables were excluded from further analysis due to extreme skewness that could lead to overfitting (Figure 2.9):

- "repaglinide", "nateglinide", "chlorpropamide"
- "acetohexamide", "tolbutamide", "acarbose", "miglitol", "troglitazone", "tolazamide"
- "examide", "citoglipton", "glyburide-metformin",
- "glipizide-metformin", "glimepiride-pioglitazone", "metformin-rosiglitazone", "metformin-pioglitazone"

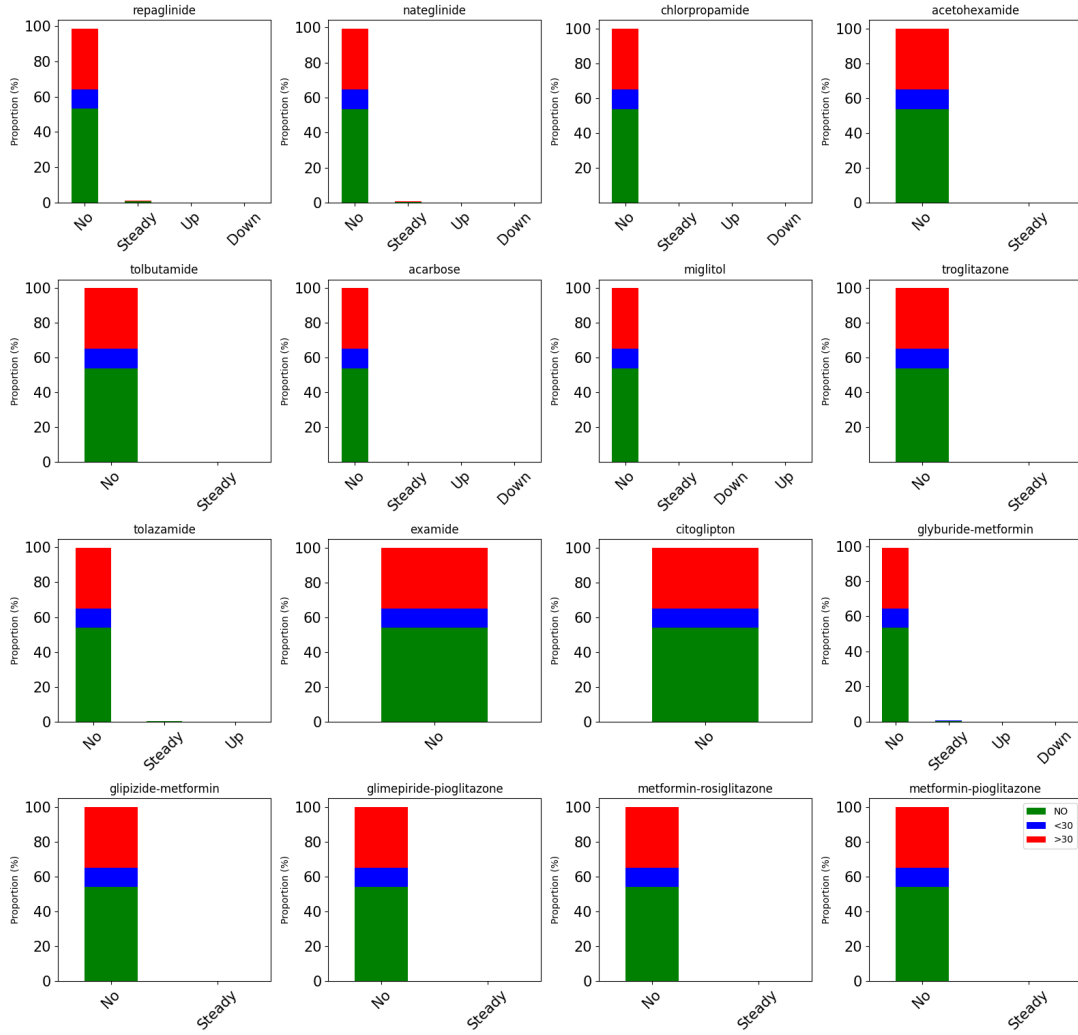


Figure 2.9: Distribution of readmission status across Other process

2.4 Numerical Variable Analysis

The numerical variables in the dataset include:

- time_in_hospital: Length of hospital stay
- num_lab_procedures: Number of laboratory procedures
- num_procedures: Number of procedures other than laboratory
- num_medications: Number of distinct medications
- number_outpatient: Number of outpatient visits
- number_emergency: Number of emergency visits
- number_inpatient: Number of inpatient visits
- number_diagnoses: Number of diagnoses
- age_mid: Midpoint of age range

2.4.1 Correlation Analysis

Correlation analysis was conducted to identify potential relationships between numerical variables. The correlation matrix did not reveal high collinearity between the variables, with most correlation coefficients below 0.4. (Figure 2.10)

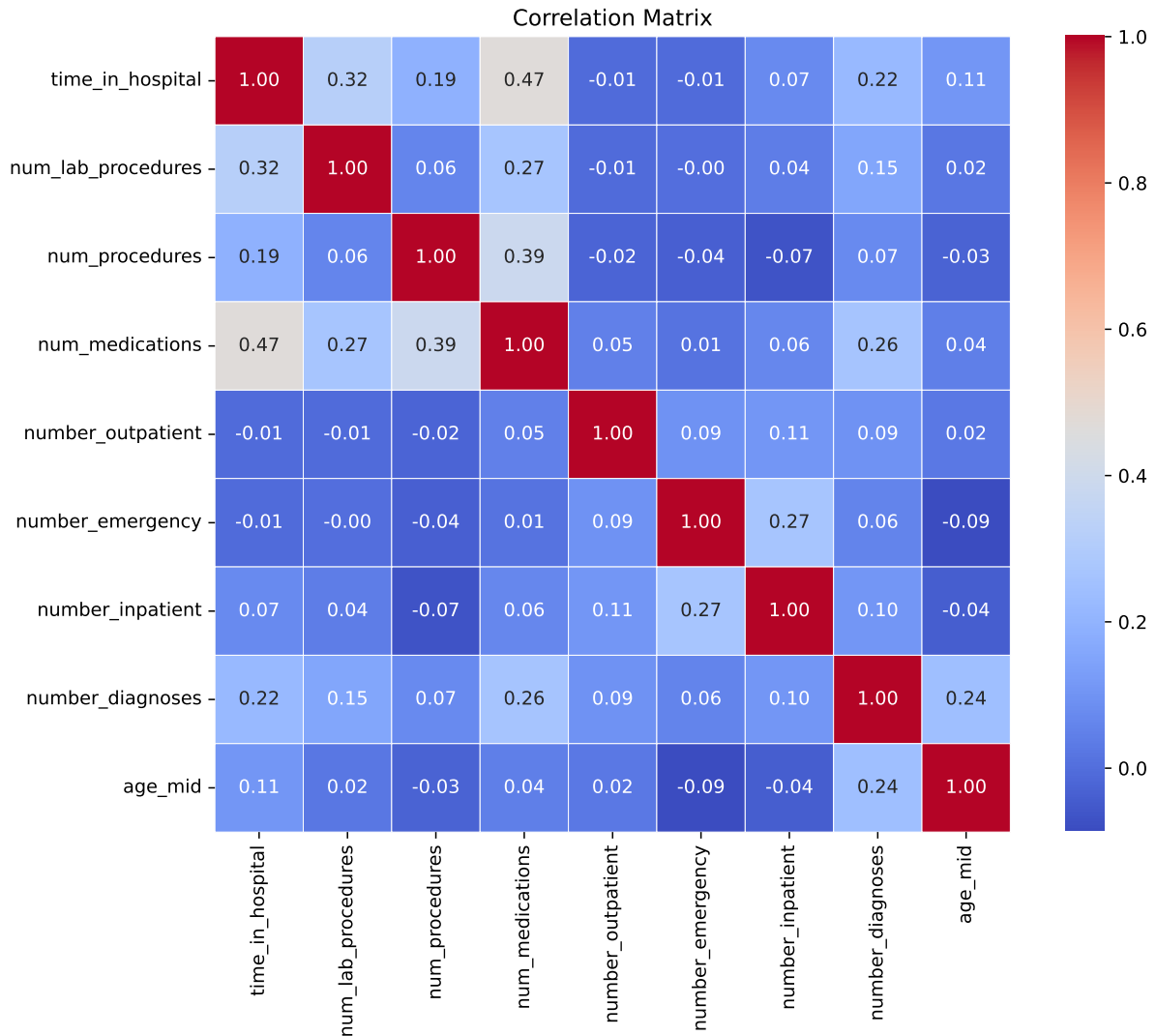


Figure 2.10: Correlation matrix of numerical variables

Variance Inflation Factor (VIF) analysis was also performed to detect multicollinearity. Although some potential collinearity was observed between "number_diagnoses" and "age", the VIF values remained within acceptable ranges (<10), and no numerical variables were removed based on this analysis. (Figure 2.11)

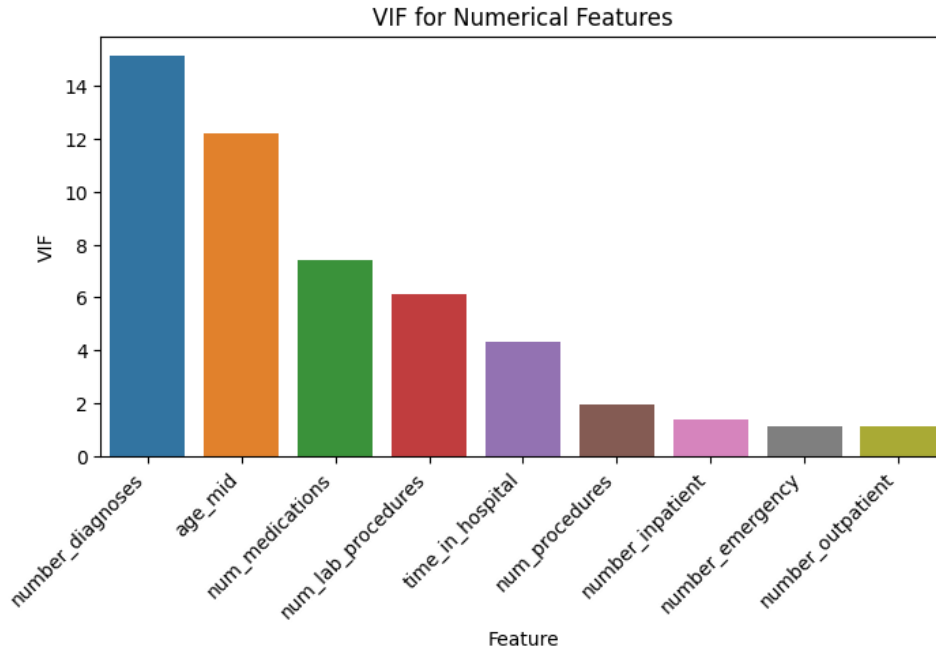


Figure 2.11: Variance Inflation Factor for numerical features

2.4.2 Distribution Analysis

In addition to the correlation and VIF analysis, distribution analysis was conducted to identify potential probability distribution issues with the numerical variables. In Figure 2.12, Histograms of numerical variables revealed:

- Gaussian distribution for num_lab_procedures
- Right-skewed distributions for time_in_hospital, num_procedures, and num_medications
- Left-skewed distribution for age_mid and number_diagnoses
- Extreme right-skew for number_outpatient, number_emergency, and number_inpatient, with most values being zero

Due to the extreme skew in outpatient, emergency, and inpatient visit counts, these variables were transformed into categorical variables:

- cat_outpatient: 0 for no visits, 1 for one or more visits
- cat_emergency: 0 for no visits, 1 for one or more visits
- cat_inpatient: 0 for no visits, 1 for one visit, 2 for more than one visit

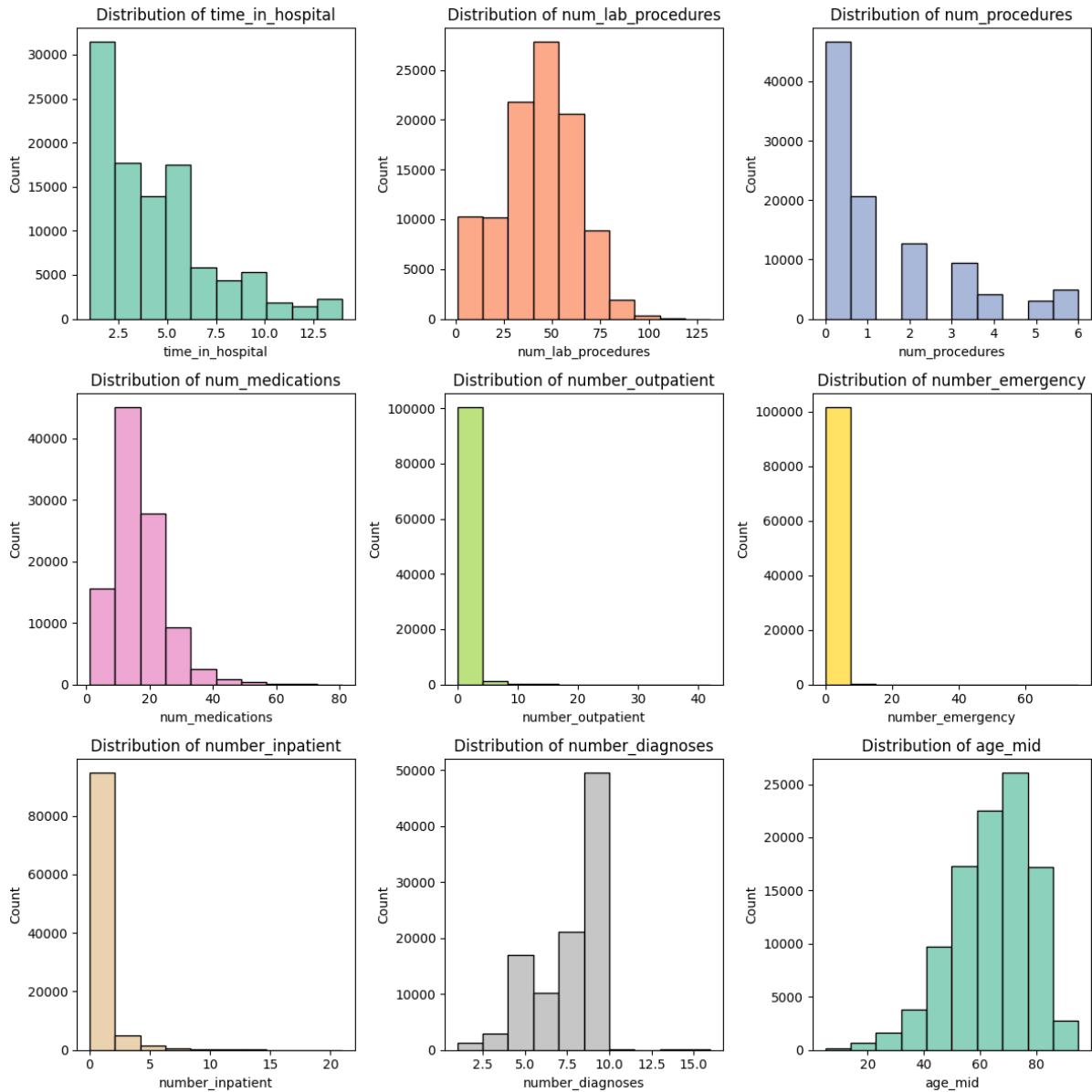


Figure 2.12: Distribution of numerical variables

2.4.3 Relationship with Target Variable

Boxplots in Figure 2.13 were used to visualize the relationship between numerical variables and the readmission status. The analysis showed comparable distributions across the three readmission categories, indicating that standardization of numerical variables would be appropriate before integration with categorical features.

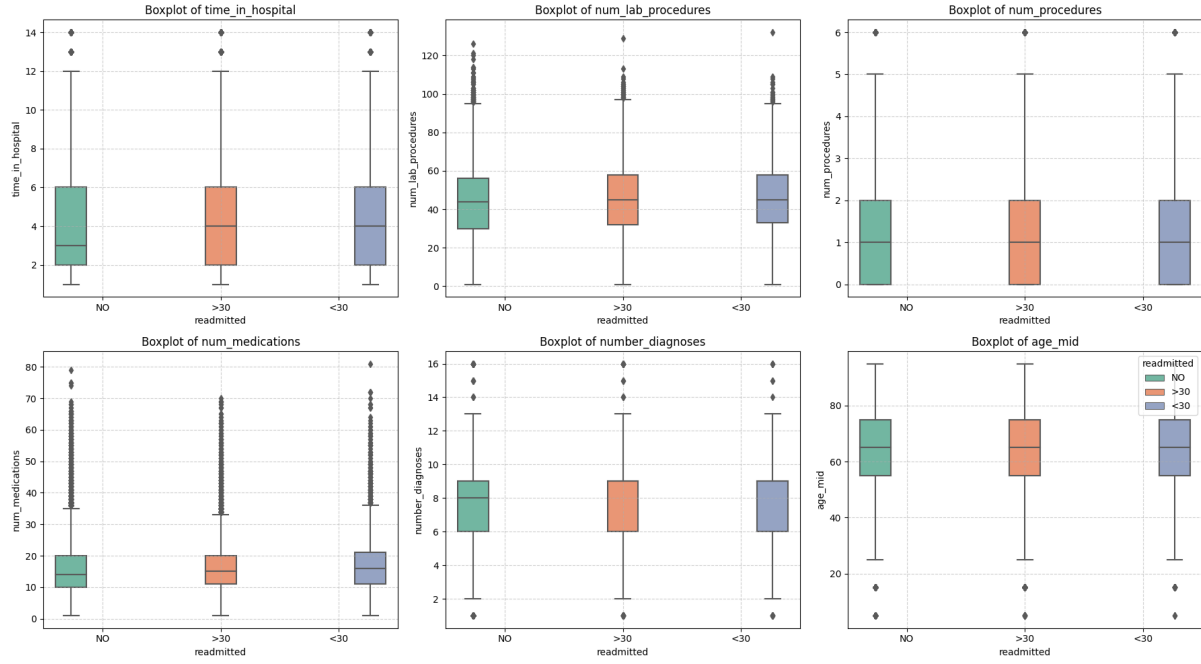


Figure 2.13: Boxplots of numerical variables by readmission status

2.5 Feature Engineering and Preparation for Modeling

After comprehensive exploratory analysis, the following preprocessing steps were implemented to prepare the data for modeling:

1. Merging of the transformed categorical variables (cat_outpatient, cat_emergency, cat_inpatient) with the selected categorical features
2. One-hot encoding of categorical variables, resulting in 193 binary features
3. Standardization of remaining numerical variables
4. Integration of encoded categorical features with standardized numerical features to form the final feature matrix X

This preprocessing pipeline transformed the original dataset with 50 variables into a modeling-ready matrix while preserving the information content and addressing the identified data quality issues.

Chapter 3

Modeling with Logistic Regression

3.1 Model Selection

Logistic regression was selected as the primary modeling approach for this analysis due to its interpretability, statistical robustness, and suitability for classification problems. As a parametric model, logistic regression provides clear insights into the relationship between predictors and the probability of hospital readmission, facilitating clinical interpretation and decision-making.

For this multiclass classification problem with three readmission categories ("NO", ">30", "<30"), we employed a multinomial logistic regression approach. This extension of the binary logistic model allows for the prediction of probabilities across multiple classes.

3.2 Data Partitioning

Prior to model training, the preprocessed dataset was partitioned into training and testing sets using a stratified random sampling approach to maintain the class distribution of the target variable. The data was split with 80% allocated to the training set and 20% to the testing set (Table 3.1).

Table 3.1: Dataset Split by Readmission Category

Subset	Size	NO	>30	<30
Training	81,413	43,891 (53.9%)	28,436 (34.9%)	9,086 (11.2%)
Testing	20,353	10,973 (53.9%)	7,109 (34.9%)	2,271 (11.2%)

3.3 K-Fold Cross-Validation

To ensure robust model evaluation and mitigate the risk of overfitting, K-fold cross-validation was implemented with $K = 10$. This procedure involved:

1. Partitioning the training data into 10 equal-sized folds
2. Training the model on 9 folds and validating on the remaining fold

3. Rotating through all folds as the validation set
4. Averaging performance metrics across all iterations
5. The entire process was executed via LogisticRegressionCV function in scikit-learn

Cross-validation provides a more reliable estimate of model performance by evaluating it on multiple subsets of the data, reducing the impact of sampling variability on performance assessment and also to avoid overfitting. Figure 3.1 shows the optimal regularization strength for each class and this regularization strength was used for the final model.

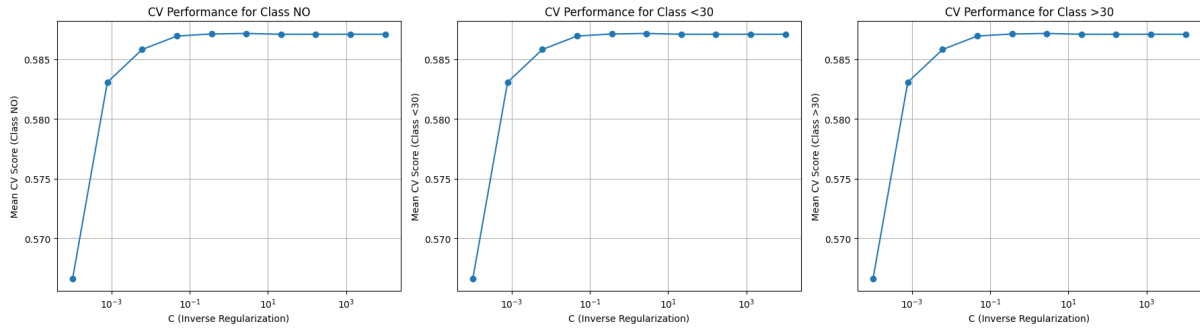


Figure 3.1: Confusion matrix for the logistic regression model on the test set

3.4 Model Training

The logistic regression model was trained using the scikit-learn implementation with the following hyperparameters:

- Regularization strength (C): was found from the cross-validation results
- Penalty: L2 (Ridge regression)
- Solver: lbfgs (Limited-memory Broyden–Fletcher–Goldfarb–Shanno algorithm)
- Multi-class strategy: multinomial
- Maximum iterations: 1000
- scoring: accuracy
- n_jobs: -1 (use all available CPU cores)

Prior to training, numerical features were standardized using `StandardScaler()` to have zero mean and unit variance, ensuring that features with larger scales did not disproportionately influence the model coefficients.

3.5 Model Evaluation

The model was evaluated on both the cross-validation folds and the held-out test set using multiple performance metrics to provide a comprehensive assessment of its predictive capabilities.

3.5.1 Performance Metrics

The following metrics were used to evaluate model performance:

- Accuracy: The proportion of correct predictions across all classes
- F1-score: The harmonic mean of precision and recall, calculated for each class and averaged. f1-score is particularly useful when dealing with imbalanced datasets as in this case.
- Confusion Matrix: A tabulation of true vs. predicted classes, revealing class-specific performance
- Classification Report: A comprehensive summary including precision, recall, and F1-score for each class

3.5.2 Test Set Results

The model's performance on the held-out test set in Table 3.5.2 and Figure 3.2:

Class	Precision	Recall	F1-score
NO	0.61	0.84	0.71
>30	0.51	0.36	0.43
<30	0.23	0.01	0.02
Accuracy	0.58		

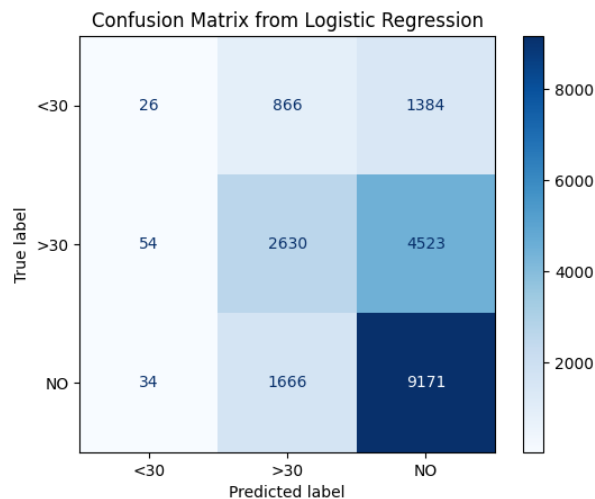


Figure 3.2: Confusion matrix for the logistic regression model on the test set

The model demonstrates stronger performance in identifying patients who will not be readmitted ("NO" class) compared to those who will be readmitted. The more challenging class to predict is readmission within 30 days ("<30"), which has the lowest recall and F1-score. This is likely due to the class imbalance in the dataset, where the "<30" class represents only 11.2% of the observations.

The ROC curves on the other hand indicate better AUC score for all 3 classes is shown in Figure 3.3:

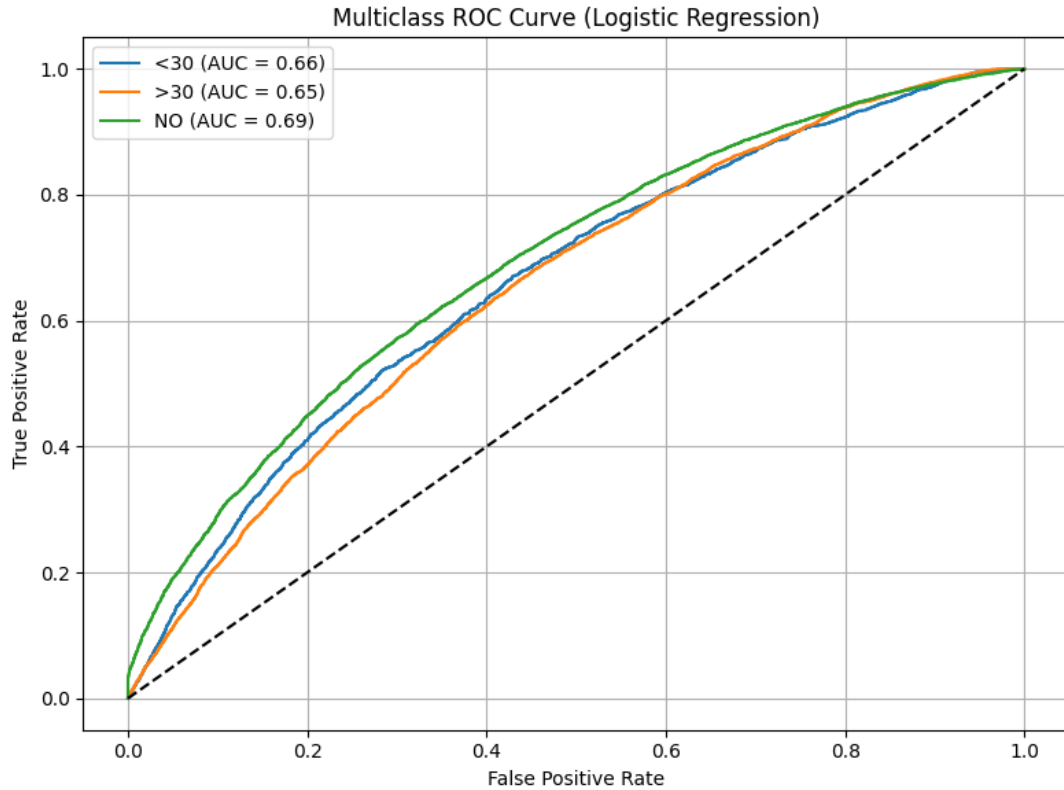


Figure 3.3: ROC curve for the logistic regression model on the test set

3.6 Feature Importance Analysis

A key advantage of logistic regression is the interpretability of model coefficients, which provide insights into the relative importance of features in predicting readmission probabilities.

3.6.1 Coefficient Interpretation from Logistic Regression

For multinomial logistic regression, the model produces a set of coefficients for each class (except the reference class). The magnitude and direction of these coefficients indicate the strength and nature of the relationship between each feature and the log-odds of belonging to a particular class (Figure 3.4).

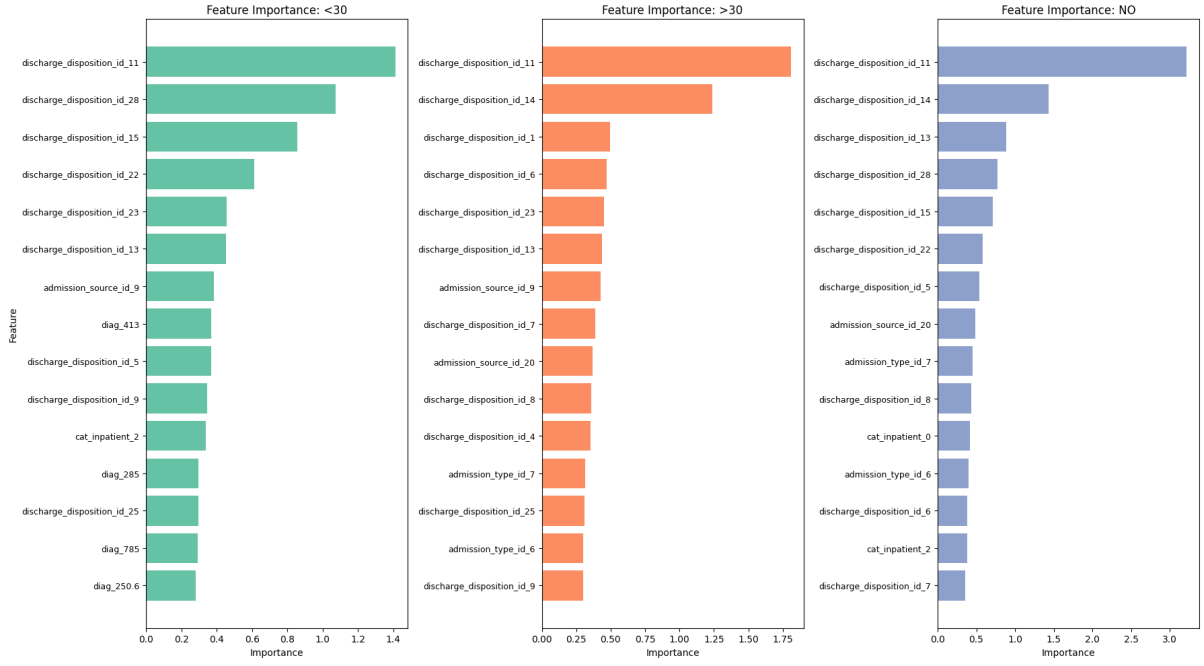


Figure 3.4: Top features by importance for each readmission category

- The top 5 features for each class are among discharge disposition id code 11, 14, 15, 28 stands for "Expired", "Hospice", "Discharge within this institution", "Discharge to psychiatric facility" respectively.

- The Top features found from Logistic Regression are based on the coefficients of the model. The Logistic Regression model is a linear model and hence the coefficients are directly proportional to the importance of the features, assuming linear relationship between the features and the log-odds of the target variable. This hence makes the categorical features like discharge disposition stronger predictor.

3.6.2 Feature Importance from Random Forest

The above analysis indicates that there exists potential non-linear relationship between the features and the target variable. Therefore, in order to find the importance of the features, it is recommended to use the Random Forest approach (Figure 3.5):

The importance features found from Random Forest model are more robust and can capture the non-linear relationship between the features and the target variable. Although the level of importance are not really strong but it is varying across predictors and provides more reliable results than the Logistic Regression model.

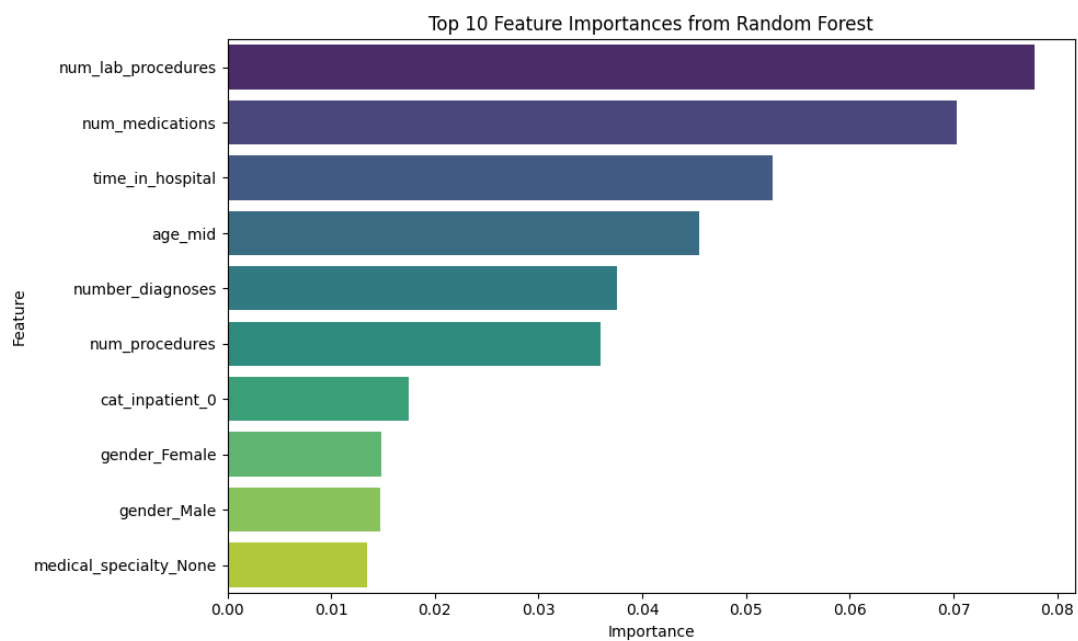


Figure 3.5: ROC curve for the logistic regression model on the test set using Random Forest

Chapter 4

Conclusions

4.1 Summary of Findings

This comprehensive analysis of diabetic patient data yielded several significant insights regarding hospital readmission patterns:

1. The exploratory data analysis revealed substantial variation in readmission rates across demographic factors, admission characteristics, and treatment regimens. The dataset exhibited class imbalance, with only 11.2% of patients readmitted within 30 days, compared to 34.9% readmitted after 30 days and 53.9% not readmitted.
2. Missing value analysis identified "weight" as the most problematic variable with 96.9% missing data, necessitating its exclusion. Other variables with notable missingness but are included since they are categorical variables and the missing values can be imputed with None.
3. Categorical variable analysis highlighted the significance of hospital administrative factors (admission type, discharge disposition, admission source) and medication regimens (particularly insulin, metformin, and glipizide) in differentiating readmission patterns.
4. Numerical variable analysis demonstrated certain weak relationships between readmission and factors such as length of hospital stay, number of diagnoses, and previous hospital encounters, though none exhibited strong individual correlations with the target variable.
5. The logistic regression model achieved moderate predictive performance with overall accuracy of 58% on the test set, demonstrating stronger capability in identifying non-readmissions than in predicting early readmissions.
6. Feature importance analysis identified discharge disposition as the most significant feature in predicting the readmission (using Logistic Regression approach).
7. The Random Forest model on the other hand identified num_lab_procedures, num_medications, age, gender and time_in_hospital as key predictors of readmission.

4.2 Methodological Limitations

Several limitations of this analysis should be acknowledged:

1. **Class Imbalance:** The significant class imbalance in the dataset, particularly the under-representation of early readmissions, posed challenges for model training and likely contributed to the lower recall for the "<30" class.
2. **Missing Data:** Despite careful imputation strategies, the high proportion of missing values in several variables, including "weight", "max_glu_serum", "A1Cresult", "medical_specialty" and "payer_code," may have introduced bias into the analysis.
3. **Limited Clinical Context:** The dataset lacks detailed clinical information such as laboratory values, vital signs, and severity scores that might enhance prediction accuracy.
4. **Temporal Aspects:** The analysis does not account for potential temporal trends or changes in clinical practice over the data collection period (1999-2008).
5. **Model Complexity:** While logistic regression offers interpretability, more complex models like Random Forest or XGBoost might capture non-linear relationships and interactions between variables more effectively.

4.3 Future Research Directions

Building on this analysis, several avenues for future research warrant exploration:

1. **Advanced Modeling Approaches:** Investigating more sophisticated modeling techniques such as ensemble methods, deep learning, or survival analysis may improve predictive performance, particularly for the challenging early readmission class.
2. **Feature Engineering:** Developing more nuanced features that capture temporal patterns of care, medication adherence, or disease progression could enhance model performance.

4.4 Conclusion

This analysis of diabetic patient data has provided valuable insights into the complex interplay of factors associated with hospital readmission. The logistic regression model developed through this work offers a foundation for identifying high-risk patients and tailoring interventions to reduce readmission rates. While the model demonstrates moderate predictive performance, its strength lies in the interpretability of its findings and their alignment with clinical understanding.

The identified risk factors—particularly discharge disposition, previous hospitalizations, multiple diagnoses, and medication changes—represent actionable targets for quality improvement initiatives. By focusing on these aspects of care, healthcare providers can work toward reducing the substantial burden of hospital readmissions among diabetic patients, ultimately improving both clinical outcomes and healthcare resource utilization.

Further refinement of predictive models and prospective evaluation of targeted interventions based on these findings will be essential to translate this analytical work into meaningful improvements in diabetic patient care.